

Aakash Kumar Nain

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Experience

Emergence.ai

Jan 2024 – Aug 2025

Senior Machine Learning Engineer (Research)

Delhi, India

Publications

- **Transformer Layers as Painters:** Published in **AAAI 2025**, this paper investigates the representational space and flow of information in frozen pretrained transformer-based models.
- **The Ungrounded Alignment:** Published in **CogSci 2025**, this paper focuses on the importance of plasticity and demonstrates a simple way to encode specific desired innate behaviors in modality-agnostic models.

Project - Image and Text Moderation

- Worked with the **Scratch** team on the image moderation model to filter out NSFW images based on the image's content, including visuals and text. The system processes 60M images/month.
- Developed synthetic data generation pipelines and helped build the internal infrastructure for training LLMs.

H2O.ai

April 2021 – Oct 2023

Senior Data Scientist

Delhi, India

- **KYC:** Worked with the CBA team and helped them leverage **Document.ai** for extracting information from KYC acceptable documents in real-time using **Intelligent Document Processing**. This automation helped CBA process 6,000 pages per hour or 0.5M pages per week, reducing existing operational costs by more than 65%.
- **Health Claims Inspection App:** Developed and deployed a deep learning-based solution for detecting fraud in health claims for IFFCO-Tokio. The system reduced operational costs by approximately 30% and processes more than 2,000 documents every day.

Ola Electric Mobility Pvt. Ltd.

July 2018 – March 2021

Research Engineer II

Bangalore, India

- **Advanced Driver Assistance System:** Developed and deployed a vision-based ADAS with a forward collision warning system in **500 OLA cabs** to prevent road accidents in India within a budget of \$200.

Paralleldots (Artzen Software Labs Pvt. Ltd.)

Jan 2017 – Sep 2017

Data Scientist

Gurgaon, India

- Co-developed the **Cornea** app, an end-to-end AI-powered photo editor created for Instagram influencers. Featured in [YourStory](#) and [Product Hunt](#) with 25,000+ downloads in the first week.

Projects

nanoGPTJAX

- Highly scalable and efficient template for implementing LLMs at scale in pure JAX without using any high-level libraries. Inspired by modded-nanogpt and nanochat for speed running pretraining runs on fineweb10B.
- Comes with code for both pretraining and post-training. Implements latest recipes like Muon optimizer, Cautious Weight Decay, RoPE, Grouped-Query Attention, and KVCache.

Equinox

- Contributed the Mistral-7B code example. Fixed [bugs](#) in normalization layers that caused instabilities when training LLMs in Equinox.

Keras 3.0 (Core Collaborator) / Keras 2.0 (Contributor)

- One of the first core collaborators in the Keras core [team](#) that implemented the multi-backend Keras.
- Contributed JAX NN ops, merging layers, metrics, loss functions, data adapters, applications, and code examples.
- Contributed the following **official code examples**: [OCR for captcha images](#), [Handwriting recognition](#), [Image captioning](#), [Denoising Diffusion Models](#), [WGAN-GP](#), and [CycleGAN](#).

TensorFlow (Contributor and Maintainer)

- Maintained and contributed to the TensorFlow-addons package for four years. Contributions include [Focal loss](#), [Cohen's Kappa](#), and [Gelu](#) activation layer.

Technical Skills

- **Programming Languages:** Python (Proficient), C/C++ (Prior exp.)
- **Machine Learning:** JAX, PyTorch, Keras, Equinox, TensorFlow, XGBoost, Numpy, OpenCV

Education

GGs Indraprastha University

2012 – 2016

Bachelor of Technology in Computer Science

Delhi, India